

C1 · From the W1 project to defensible evidence

Business Intelligence · IIB423T-1

Universidad del Desarrollo · Facultad de Ingeniería

Instructor: Tomás Fontecilla Correa · Tomas.fontecilla@udd.cl

Assessment information	Details
Submission	Tuesday 29 September 2026, before 17:50
Presentation and individual defense	29 September, 17:50-20:20 · Y035
Time zone	Santiago, Chile
Teams	Same team as W1; pairs preferred; authorized individual work remains available
Weight	30% of the pre-exam course grade (NP)
Assessment structure	60 points: team submission · 40 points: individual defense
Tools	Python/Jupyter recommended; other verifiable tools accepted; GitHub optional

Your challenge: develop the line selected in W1 into a defensible C1 analysis. Show how your data preparation and EDA support the question and a clearly bounded decision.

1. C1 develops the project you selected in W1

In W1, your team explored candidate questions with public data and selected one line to continue. C1 develops that line into a complete, defensible analysis. Your question, KPI or measure, data preparation and EDA must build directly on the evidence, decisions and feedback documented in W1.

Professional analytical work includes deciding what deserves investigation, distinguishing a plausible explanation from evidence, and translating a broad concern into measurable events and indicators. A step-by-step recipe would make these decisions for you. In this project, those decisions are part of what you must demonstrate.

AI can help you search, write code, transform data and communicate results. It cannot take responsibility for an unsupported claim, an invalid denominator or a misleading join. You remain responsible for checking sources, calculations and interpretations. A justified change of direction can be stronger evidence of learning than defending your first idea at any cost.

This is an educational observatory, not a commissioned clinical study. No interviews or external stakeholder validation are required. Use literature, technical reports and official health priorities to establish context. Identify a plausible user and decision as your proposal; do not invent what patients or managers think.

2. Expected learning outcomes

The following are English renderings of the approved course outcomes. C1 assesses the stated components at the level taught so far.

Course learning outcome	What you will demonstrate in C1	Rubric
LO1. Analyze complex organizational problems and integrate information from different areas to formulate solutions.	Justify and delimit the project selected in W1; connect documentary evidence, a proposed decision and measurable questions; explain how the line evolved after W1.	C1-A, C1-B, C1-F
LO2. Use digital technologies and computational techniques to process, model and represent data in organizational contexts.	Demonstrate the processing and representation components through profiling, preparation and a reproducible analysis. Predictive modeling is outside C1.	C1-C, C1-E, C1-G
LO3. Interpret patterns, trends and relationships in data to support informed decisions.	Interpret exploratory results, assess alternative explanations and communicate decision-relevant conclusions with limits.	C1-D, C1-H

The scope is Unit I plus data exploration, profiling and cleaning from Unit II. Machine learning, causal identification and a finished dashboard are not required.

3. Your investigation

A. Explain the selected problem and how it evolved

Identify the problem, population or service, place, period and proposed use of the analysis. Support its relevance with retrievable literature or official technical evidence. Explain its relationship to a relevant MINSAL priority or plan where applicable; if there is no direct policy match, explain that boundary instead of inventing one. Record publication dates and versions. Distinguish a policy objective from a numerical target and from an indicator that your data can actually measure.

Summarize the alternatives considered in W1, why you selected this line, and what changed after instructor feedback. Record the advice received, your response and the resulting change or justified decision to retain the original approach. If feedback was not available, say so, identify the outstanding question and explain your provisional choice; its absence is not a student failure.

B. Apply the seven steps from Class 03

Use these names to make your reasoning easy to locate. They can appear in a short table with links to the analysis; do not repeat material unnecessarily.

Step	Evidence to provide
1. Decision	Who could use the analysis, what decision they face and what alternatives exist. Clearly label the proposed user and decision.
2. Success criteria	What improvement would mean; distinguish the desired outcome from what can be measured now.
3. Business questions	A bounded main question and any supporting questions needed to investigate it.
4. Business process	The relevant process or service pathway, including boundaries and events the data do not capture.
5. Entities, events and states	What is being counted or observed; distinguish people, encounters, establishments, activities and aggregate records as appropriate.
6. Data representation	What one row represents, identifiers, fields and relationships. Explain source coverage and any proposed or actual joins.
7. Measurement	Definitions, formulas, units, numerator and denominator where applicable, period, exclusions and interpretation limits.

C. Profile, prepare and explore the data

Start from source files and their dictionaries. Show the relevant structure, types, coverage, candidate keys, duplicate checks, missing or invalid values and distributions. Distinguish missing records from recorded zeros. Document filters, transformations and their effect on the analyzed population. Keep originals unchanged.

Use summaries, comparisons and visualizations that address your question. Explain the meaning of each important result and how it changes your understanding. When comparing periods or groups, check compatible definitions, reporting completeness and denominators. A temporal comparison is appropriate when the question and coverage support it; the use of all years or all available datasets is not a requirement.

For a join, document the key, expected cardinality, coverage of matches, unmatched records and row/count changes. Aggregate to compatible units when needed. If no join is useful, justify the choice; a forced merge earns no credit. A location of care is not automatically a patient's residence or a valid population denominator.

Apply the five lenses below to your actual investigation. State when a lens is not applicable and why.

Five C	What to demonstrate
Clean	Check validity and duplicates; justify corrections, exclusions or retention.
Consistent	Check whether definitions and calculation rules remain stable within a source across the periods studied.
Conformed	Check whether shared entities, events, codes and reference definitions agree across sources you combine.
Current	Explain whether the age and availability of the information suit the proposed decision window. Historical analysis can be appropriate.
Comprehensive	Assess coverage of the relevant population, fields, periods and exceptions, including what remains unobserved.

If there are no nulls, report the check and move on. If no cleaning is needed, justify retaining the data. Do not invent defects, imputations or transformations to appear more sophisticated. An absence of a clear pattern is an acceptable result when investigated and interpreted properly.

D. Conclude within the evidence

Answer the question as far as the evidence allows. Separate observed patterns, interpretations and hypotheses. Consider relevant alternative explanations and explain what the analysis cannot establish. Counts alone do not demonstrate risk, service quality or causal effects.

State the implications for the proposed decision and what further evidence would be needed. Connect these findings to the feasible continuation described below. Feasibility matters more than promising every possible technique.

E. Project continuity: beyond C1

Your selected project must have a **plausible path toward machine learning and a dashboard**, since these are part of the work that follows. This is a condition of project viability, not a requirement to implement either one in C1. Explain why the selected line can continue, taking into account the alternatives and instructor feedback from W1.

Describe a concrete possible future ML task and its decision value: for example, an outcome/event to predict or classify, or useful profiles to explore through clustering. Identify the unit of analysis, available inputs and any additional data needed. For a supervised task, identify the proposed outcome; do not invent outcome labels for an exploratory clustering task. If the proposed use is prospective, distinguish information available at the time of the decision from information recorded only after the outcome. Flag uncertainties to review in the next stage rather than pretending that feasibility is already proven.

For the dashboard, identify its proposed users, the decisions it would support, core indicators and useful comparisons or filters. Explain how the EDA findings inform this proposal. A dashboard is more than a gallery of charts; the intended use should guide what is shown.

Do not train a model, choose a required algorithm, design a complete validation protocol, promise predictive performance or build the dashboard for C1. If a critical part depends on unavailable data, explain the gap and propose a feasible adjustment for instructor review. A vague promise to “add ML later” does not establish viability.

4. Submission and defense

One archive with a mandatory internal structure

Submit **one ZIP or RAR file** through the designated Canvas assignment. Use `GROUP_XX_C1.zip` or `GROUP_XX_C1.rar`, where XX is your two-digit group ID. For example: `GROUP_01_C1.rar`. Keep the same group ID in W1 and C1; use the actual archive format, not a renamed extension. In the generic pattern `GROUP_XX_ASSESSMENT`, replace `ASSESSMENT` with `W1` or `C1`.

The archive must contain a single root folder with the same name, without the extension. The following structure and the report filename are mandatory; replace 01 with your group ID. Choose descriptive filenames inside `analysis/` and `data/`.

```
GROUP_01_C1/  
  README.md  
  report/  
    GROUP_01_C1_Report.pdf  
  analysis/  
    [editable notebooks, scripts, workbooks or queries]  
  data/  
    [reduced input data and generated case tables]  
  presentation/  
    GROUP_01_C1_Slides.pdf
```

The report PDF is mandatory and self-contained. The report must contain the selected problem and its evolution from W1, documentary support, the seven-step map, data/quality decisions and Five Cs, main EDA results with interpretation, limitations and future ML/dashboard feasibility, followed by the Process record and references. The instructor must be able to understand the complete investigation from this PDF alone, without opening a notebook, workbook or the project. Explain the analytical method and consequential choices, show the tables/figures needed to support the conclusions with units and context, and discuss results and limitations. “See the notebook” does not replace those explanations or the evidence needed to follow the argument. Technical files support verification and reproduction. Keep notebooks well documented, but do not paste every code cell into the PDF. The presentation PDF is also mandatory and does not replace the report; include its editable source when available, but a Gamma or other online link does not replace the slides PDF.

Submit executed notebooks with saved outputs. Before submission, restart the kernel, run the analysis from top to bottom in the documented order, and save the `.ipynb` with its tables, figures and relevant outputs visible. Outputs must match the submitted code and data. Do not clear them or mix stale results from earlier versions. If an unresolved error prevents a part from running, identify that section and its effect on the findings in the PDF and README; do not present missing or unverified execution as completed. For Excel, Power Query or equivalent tools, save the calculated/refreshed results together with formulas and transformation steps. The reviewer must be able to inspect the submitted results before deciding whether a rerun is needed.

README.md is the package entry point. Identify members and group ID, submission date/version, file roles and the order in which to open/run/refresh them. State the software and versions used, required libraries or connectors and how to install/configure the dependencies. For Python, a requirements file may complement these instructions. Explain paths, source versions and any unresolved execution issue. Use project-relative paths rather than paths tied to your own computer.

Include the project files and the data for your reduced case. In `data/`, include the relevant smaller inputs or an uncleaned extract for the selected line, as well as prepared tables needed to verify the reported results. Label which files are extracted inputs and which are processed outputs; document their source, scope and transformations. The editable analysis must show how these inputs became the analyzed dataset and results. If retaining valid data without changes was justified, identify that explicitly instead of creating unnecessary copies or cleaning steps.

You may omit giant original datasets, but the README must identify the exact public source or course file/version, retrieval instructions and the script/query or documented filters that reconstruct the reduced input. Where downloads can change, record the version/date used and retain your local original. The reviewer should be able to inspect the reduced case without downloading the entire collection and trace how that case was obtained. A final summary table alone is insufficient if its preparation cannot be checked.

Include the relevant editable files, saved outputs and available source documents used to prepare the delivery; do not include installed applications, Python installations, `.venv/Conda` environment folders, caches, temporary files or credentials. Describe dependencies instead. This is a project snapshot, not a copy of the entire computer. Check that the archive opens and that its internal paths work after extraction.

Process record: a required section of the report PDF

Place a section titled **Process record** after the analytical brief and before the references. It explains how the work developed; it is not a dump of code, chats or every command executed. Use the following subsections and concise tables where helpful:

Subsection	What to record
Decisions and checks	Date/stage; consequential decision; alternative considered; reason and supporting evidence; check performed; result or change; location in the submitted files.
Instructor feedback and response	Advice received and when; whether you accepted, adapted or rejected it; your reason; change made and evidence location. If feedback is pending, state that and your provisional decision.
Individual contributions	Member; concrete contribution; location of evidence. Record actual contributions, not only intended task allocation. Equal volume is not required.
AI use and verification	Tool and purpose; relevant prompt/output excerpts; what you accepted, modified or rejected; how you checked consequential suggestions. State explicitly if no AI was used.
Individual reflection	Each member identifies a contribution, a learning gain and one unresolved uncertainty. An authorized solo submission follows the same structure.

For example, a decision entry can follow: **Question/issue; alternatives; decision and reason; verification; evidence location**. Cite a notebook heading, workbook sheet/range, query or file path so the reviewer can inspect the evidence. Your PDF must contain the reasoning itself; links to notebooks or external chats cannot replace it. References, code comments and output explanations complement this record rather than substitute for it.

Python/Jupyter is the teaching environment. Excel, Power Query and other tools receive the same assessment when their procedures and outputs can be inspected. Test your notebook from top to bottom, refresh your queries, or verify workbook formulas and dependencies, as appropriate. Report any unresolved reproducibility issue rather than hiding it.

AI use is allowed during preparation. Identify the tool, purpose, relevant prompts/output excerpts and what you accepted, changed or rejected; connect consequential suggestions to your verification. If no AI was used, state that. Use public data and do not upload confidential information. You must be able to explain the submitted work. No live AI consultation is allowed during the defense.

The scheduled slot is up to **25 minutes per project including questions and transitions**; the instructor will communicate the order. Both members must answer individual questions. An authorized solo project follows the same criteria. Institutional accommodations and existing course rules apply; no new late-submission rule is introduced for C1 here.

Contributions need not be equal in volume; document what each person did. Unequal volume alone does not reduce the shared score. Individual understanding is assessed through each student's defense, not inferred from the amount of code they wrote. Each student will have an opportunity to address F, G and H; unobserved performance is recorded as pending until the instructor has evidence.

5. How points are assigned

The submission receives a shared score out of 60. Each student receives their own defense score out of 40. **Individual C1 points = shared submission points + that student's defense points**. There is no second 60/40 multiplication. Institutional conversion to grades is applied by the instructor.

For each criterion, the reviewer selects one published level and its fixed points: Fully achieved = 100% of its weight; Mostly achieved = 80%; Partially achieved = 40%; Not achieved = 0%. No intermediate scores are interpolated. All published points are whole numbers. Add them directly; no rounding is required.

The **core evidence** under each criterion is necessary for Fully or Mostly achieved. Fully requires all the listed dimensions to be met; Mostly permits only a minor omission that does not undermine the core evidence. Relevant but incomplete or materially flawed evidence is Partially achieved. Not achieved means no assessable demonstration of that criterion. Borderline cases must be explained against the descriptors and resolved by the instructor.

The future ML/dashboard proposal is scored under **C1-D only**. C1-A assesses the documented problem, selection and response to feedback; the same missing or weak future proposal must not also lower C1-A.

No deductions are added outside this rubric for the same analytical defect. Where a defect affects separate criteria, the reviewer must explain the distinct affected performances. Tool choice, length, number of charts, null-free data, justified changes after feedback and appropriately inconclusive results do not reduce points. A technical inability to inspect a file is flagged for clarification, not automatically treated as absent work. Assisted review may support the instructor; all final assessment decisions remain with the instructor.

C1-A · Problem, foundation and evolution from W1 — 15 points

Core evidence: a bounded, document-supported problem and a reasoned account of the selection from W1.

Level	Points	Descriptor
Fully achieved	15	Delimits the problem and intended use; uses relevant, retrievable evidence; distinguishes policy context from measurable claims; compares the prior options and explains the response to available feedback and subsequent changes.
Mostly achieved	12	The bounded problem, supporting evidence and reasoned selection are clear. A contextual detail or part of the change record is incomplete, without altering the justification; available feedback is considered.
Partially achieved	6	Identifies a relevant topic or some supporting evidence, but the problem is broad, key claims lack support, or the W1 selection/feedback response is unexplained.
Not achieved	0	Provides neither an identifiable W1 project/problem nor relevant documentary evidence or reasoning for investigating it.

C1-B · Questions, metrics and data fit — 10 points

Core evidence: a main question linked to the proposed decision and at least one interpretable measure grounded in the actual unit of observation.

Level	Points	Descriptor
Fully achieved	10	Applies all seven steps coherently. Entities, events, rows and process boundaries align with the question; metric formulas, units, scope and denominators/exclusions are explicit; proposed conclusions stay within what the sources observe.
Mostly achieved	8	The decision, question, unit of observation and metric definitions form a valid chain. A supporting process detail or nonessential step explanation is incomplete, without compromising measurement.
Partially achieved	4	Provides relevant questions or calculations, but confuses observation units, leaves a key measure undefined, or claims the data capture events they do not observe.
Not achieved	0	Provides neither a relevant analytical question nor an assessable metric definition/calculation or question-to-data mapping.

C1-C · Profiling, preparation and the Five Cs — 15 points

Core evidence: observable quality/coverage checks and justified handling of issues that could affect the analysis, including join validation when used.

Level	Points	Descriptor
Fully achieved	15	Profiles relevant fields, keys and coverage; applies each C or justifies non-applicability; records and validates transformations or justified retention. Used joins have cardinality, match and row/count checks. Material unresolved limitations are identified and the scope adjusted accordingly.
Mostly achieved	12	Core checks support a defensible prepared dataset, with material risks handled and used joins validated. A secondary check or a nonessential before/after detail is incomplete.
Partially achieved	6	Shows actual profiling or preparation, but a material issue remains unexamined, an alteration lacks justification, or a used join lacks validation that could reveal duplication or lost coverage.
Not achieved	0	Provides neither actual profiling/quality checks nor assessable reasoning for preparing or retaining the analyzed data.

C1-D · EDA, interpretation and limitations — 15 points

Core evidence: inspectable exploratory results relevant to the question, with interpretations grounded in those results.

Level	Points	Descriptor
Fully achieved	15	Uses suitable summaries/comparisons and visuals; interprets results with comparability checks and limits; distinguishes observation from explanation; connects findings to a concrete future ML task, available/missing data and useful dashboard users/indicators. No implementation is required.
Mostly achieved	12	Relevant results support the interpretation and its limits, with an identifiable future ML task and dashboard use grounded in the findings. A secondary comparison, labeling or future data/filter detail is incomplete.
Partially achieved	6	Shows relevant outputs but omits interpretation or mainly describes charts; uses misleading comparisons or unsupported conclusions; or offers only generic ML/dashboard promises without an identifiable task or decision use.
Not achieved	0	Provides neither inspectable exploratory results relevant to the question nor a relevant interpretation grounded in identifiable analytical evidence.

C1-E · Reproducibility, contributions and AI accountability — 5 points

Core evidence: an inspectable route from sources to reported results, identified contributions and an explicit AI-use statement.

Level	Points	Descriptor
Fully achieved	5	The prescribed archive contains a report PDF understandable without the project, its Process record, slides PDF, executed/saved analysis with outputs, reduced-case inputs/outputs and usable README; versions, decisions and contributions are traceable, with self-assessments and relevant AI checks.
Mostly achieved	4	The report explains the investigation on its own; required PDFs, process evidence, saved analysis outputs and reduced data support verification; contributions and AI use are explicit. A minor naming/navigation detail or nonessential process entry is incomplete.
Partially achieved	2	Provides some inspectable work, but a required PDF or Process record is absent, the report depends on external files for understanding, or required saved outputs, reduced inputs or editable analysis are missing; alternatively, reproducibility, contributions or AI accountability have material gaps.
Not achieved	0	Provides no usable analytical/process trail beyond unsupported final outputs.

C1-F · Individual understanding of the problem and data — 15 points

Core evidence: the student's own explanation of the problem, analytical scope and what the data represent.

Level	Points	Descriptor
Fully achieved	15	Independently explains the relevance and selection of the problem, the observation unit and source coverage, and connects these to the decision and measurement limits.
Mostly achieved	12	Independently explains the core problem and data correctly; needs clarification of a secondary contextual detail, without losing the argument.
Partially achieved	6	Explains some elements but relies on memorized labels or the partner, or materially confuses units, coverage or purpose.
Not achieved	0	After a fair opportunity to answer, demonstrates no assessable understanding of the submitted problem and data.

C1-G · Individual justification of analytical decisions — 15 points

Core evidence: the student's own explanation of a consequential method or preparation choice, its reason and a relevant check.

Level	Points	Descriptor
Fully achieved	15	Justifies a consequential choice with evidence, compares a plausible alternative, explains the validation used and identifies how a changed assumption would affect the analysis.
Mostly achieved	12	Correctly justifies the consequential choice and its validation; discussion of a plausible alternative or of a changed assumption is missing or incomplete, without undermining the explanation of the chosen approach.
Partially achieved	6	Describes operations but cannot justify an important choice or check, or relies on AI/tool authority as the reason it must be correct.
Not achieved	0	After a fair opportunity to answer, provides no assessable explanation of any consequential analytical decision.

C1-H · Individual interpretation and responses — 10 points

Core evidence: the student's own interpretation of a reported result and a relevant limit on its use.

Level	Points	Descriptor
Fully achieved	10	Explains the result, distinguishes evidence from a hypothesis, responds to a challenge using the analysis, and identifies a justified implication or further evidence needed.
Mostly achieved	8	Correctly interprets the result and its main limit and responds to the central question; a secondary implication or alternative explanation is incomplete.
Partially achieved	4	Recites results but cannot connect them to the question, overlooks a material limit, or maintains an unsupported inference when challenged.
Not achieved	0	After a fair opportunity to answer, provides no assessable interpretation of the results.

6. Sources and preparation checklist

Start with the course data and dictionaries, the [DEIS portal](#), the [MINSAL open-data catalog](#), and the [FONASA GRD dashboard](#). Public data you locate independently are welcome when documented and suitable. A dashboard is not automatically a downloadable analytical table.

The [MINSAL National Health Strategy toward 2030, 2022 edition](#) is a starting point for policy context, not proof of current performance or a substitute for the relevant indicator definition. Consult applicable updates for your chosen topic. Use retrievable references with titles, institutions/authors, dates and links; connect them to the claims they support.

Before submitting, confirm that the question, metrics, units and conclusions agree; that analytical steps and data versions are traceable; that limitations and feedback responses are visible; and that each member can explain the work. No fixed number of pages, charts or citations replaces those requirements.